

Early Forest Fire Detection Using Satellite Imagery

1. Introduction

Forest fires cause severe environmental, ecological, and economic damage globally. Early detection is critical to reduce fire spread and improve emergency response efficiency. Satellite-based monitoring systems such as MODIS (Moderate Resolution Imaging Spectroradiometer) and VIIRS (Visible Infrared Imaging Radiometer Suite) provide large-scale Earth observation data that can be leveraged for early wildfire detection using Artificial Intelligence and Deep Learning techniques.

2. Traditional Approaches

Traditional wildfire detection methods relied on threshold-based thermal anomaly detection, spectral indices such as NDVI and NBR, and rule-based hotspot extraction. While effective in some cases, these approaches suffer from high false alarm rates due to clouds, sun-glint, and industrial heat sources. They also lack temporal modeling capabilities for early-stage ignition detection.

3. Deep Learning-Based Approaches

Recent research applies Convolutional Neural Networks (CNNs) for spatial feature extraction from multispectral satellite imagery. CNNs are effective at detecting fire patterns but do not inherently model temporal fire progression.

CNN + LSTM hybrid models combine spatial feature extraction with temporal modeling, improving near-real-time detection by learning fire spread dynamics across multiple time frames.

Vision Transformers (ViT) and Swin Transformers represent the state-of-the-art in wildfire detection. They capture long-range spatial dependencies and multi-scale contextual information more effectively than traditional CNN architectures, often outperforming them in complex wildfire imagery scenarios.

4. Research Gap

Despite advancements, several research gaps remain: - Limited fusion of MODIS and VIIRS datasets for improved accuracy. - Insufficient focus on early ignition-stage detection. - Limited transformer-based temporal modeling approaches. - Challenges in reducing detection latency while maintaining high precision.

5. Proposed Methodology

This project proposes a hybrid deep learning architecture combining multi-sensor data (MODIS + VIIRS) with a Vision Transformer backbone and temporal LSTM modeling. The system aims to improve early-stage wildfire detection accuracy while minimizing false alarms and detection latency.

6. Conclusion

Deep learning techniques significantly enhance the capability of satellite-based wildfire detection systems. By integrating spatial and temporal modeling and leveraging modern transformer architectures, early detection systems can become more robust, accurate, and suitable for real-time deployment.